

ElasticNet Regression Interview Questions & Answers

For Data Scientist, Junior ML Engineer, and Data Analyst interviews.

Q1. What is ElasticNet Regression?

Ridge Uses L2 regularization. Its main effect is to shrink coefficients toward zero. But generally it doesn't make them exactly zero.

$$\text{Loss} + \alpha \sum \beta_j^2$$

Lasso Uses **L1 regularization**. Its main effects is to **shrink coefficients + potentially make some exactly zero**. Therefore, Lasso can perform feature selection.

$$\text{Loss} + \alpha \sum |\beta_j|$$

Therefore, Elastic Net combines both: $\text{Loss} + \alpha [l1_ratio \sum |\beta_j| + (1-l1_ratio) \sum \beta_j^2]$

So you can think: Elastic Net=Lasso+Ridge

Not literally adding two separately trained models, but combining their **L1 and L2 penalties in one objective function**.

Q2. What does L1_ratio mean?

l1_ratio controls how much L1 vs L2 regularization is used.

The relationship is:

L1 contribution = L1_ratio

L2 contribution=1-L1_ratio

if 100%L1+0%L2 then Elastic Net becomes essentially **Lasso**.

if 0%L1+100%L2 then Elastic Net becomes essentially **Ridge**.

Q3. Then what does alpha do?

alpha: Controls the overall strength of regularization. **alpha** = How strongly should I regularize?

l1_ratio: Controls the type/mix of regularization. **l1_ratio** = What kind of regularization should I use?

Q4. What are we trying to find?

For Elastic Net, we want to find **alpha** and **L1_ratio** that gives the best validation performance.

What is Elastic Net and why would you use it instead of Lasso?"

Elastic Net combines L1 and L2 regularization. L1 encourages sparsity and can perform feature selection by making coefficients exactly zero, while L2 helps stabilize the model, particularly when predictors are correlated. The l1_ratio controls the balance between L1 and L2, while alpha controls the overall regularization strength. I would typically tune both using cross-validation, for example with GridSearchCV or RandomizedSearchCV.